

Two Regimes, One Equation: An Amplification-Factor / Spalart–Allmaras Closure for Transition-Aware RANS

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Abstract

Every modern RANS transition closure — γ - Re_θ , AFT, SA-BCM — adds at least one transport equation to the turbulence model it accompanies. We show that the extra equation is not needed. The small range of the Spalart–Allmaras (SA) working variable $\tilde{\nu}$, which the original SA model leaves essentially inert, can be reinterpreted as the amplification envelope of a Tollmien–Schlichting wave; with one algebraic modification of the SA source term, a single transport equation then carries *both* regimes — an Amplification-Factor-Transport (AFT) growth law where $\tilde{\nu} \lesssim O(\nu)$, and standard SA turbulent mixing where $\tilde{\nu} \gg \nu$. There is no auxiliary intermittency, no second transported variable, and no extra Jacobian block in an implicit solver: the entire transition–turbulence problem is closed by exactly one PDE. A novel local sensor — the *Escape Reynolds number* $Re_{\text{esc}} = yv/\nu$ — is introduced as a robust, integral surrogate for the pressure-gradient parameter λ_θ , letting a single source term span attached Tollmien–Schlichting amplification, separation-induced (Kelvin–Helmholtz) breakdown, and fully turbulent eddy-viscosity production. The closure is plug-compatible with any existing SA-based code, recovers the classical e^N envelope on a Blasius layer, and retains the standard SA behavior at fully turbulent conditions. Preliminary 1D and 2D results calibrate the growth law against $dN/dRe_\theta \approx 0.007$ and reproduce flat-plate transition trends across freestream turbulence levels. The final paper will report 2D NACA airfoil predictions of C_p , C_f , and transition location compared with XFOIL and the SA-BCM transition family, demonstrating that a single carefully constructed transport equation can carry both transition and turbulence physics with accuracy comparable to multi-equation RANS transition models, while preserving the robustness, low cost, and implementation simplicity that have made SA ubiquitous in aerospace CFD.

1 Motivation and Purpose

Transition prediction remains one of the largest sources of uncertainty in industrial RANS simulation. Most modern transition closures — the γ - Re_θ model of Langtry & Menter, the AFT model of Coder & Maughmer,

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and the SA-BCM family of correlation-based variants — introduce one or more additional transport equations and a network of empirical functions to switch between regimes. Each new state variable adds modeling surface area: tunable coefficients, limiters, and free-stream ambiguities that must be re-validated for every new geometry. From a software engineering standpoint each new variable also doubles the burden of implementation, linearization, and Jacobian-coloring inside an implicit solver.

By contrast, the original SA model has only one transported quantity, and yet remains the default closure across much of aerospace CFD. Its near-wall behavior, free-shear behavior, and zero-time-derivative response in homogeneous isotropic turbulence are not “correct” in any detailed sense, but they are robust, predictable, and easy to debug in flows the model’s authors never anticipated. There is a parsimony in this design that has proven remarkably durable.

The objective of this paper is to show that the same parsimony can be extended to laminar-to-turbulent transition. Rather than carry a separate amplification factor or intermittency, we reinterpret the small- $\tilde{\nu}$ range of the SA equation — the regime in which the standard model contributes negligibly to the eddy viscosity — as the natural state space for a Tollmien–Schlichting (TS) envelope. The AFT physics is folded into the production and destruction terms of the SA equation itself, and the closure recovers each individual model in its respective limit.

2 Technical Foundation

2.1 One transport equation, two regimes

The model retains the SA conservation form,

$$\frac{D\tilde{\nu}}{Dt} = P(\tilde{\nu}, \nabla u, d) - D(\tilde{\nu}, d) + T(\tilde{\nu}, \nabla u) + \frac{1}{\sigma} \nabla \cdot [(\nu + \tilde{\nu}) \nabla \tilde{\nu}] + \frac{c_{b2}}{\sigma} |\nabla \tilde{\nu}|^2, \quad (1)$$

in which the diffusion, near-wall destruction, and gradient-energy structure are inherited from the original Spalart–Allmaras formulation. The novelty sits in the transition-active production term T , designed so that (1) reduces to standard SA whenever $\tilde{\nu}$ is large relative to the molecular viscosity, and to an AFT amplification equation whenever $\tilde{\nu}$ is small.

2.2 A local replacement for Re_θ

In the classical e^N method, the log-amplitude of the most unstable TS wave grows almost linearly with the momentum-thickness Reynolds number, $dN/dRe_\theta \approx 0.007$ for a Blasius layer. Adopting this rule inside a RANS code is awkward because Re_θ is non-local: it requires integrating the velocity profile from the wall to a boundary-layer edge that may not be cleanly defined in 2D/3D RANS fields. We instead use the *vorticity Reynolds number*

$$Re_\Omega(x, y) = \frac{y^2 |\Omega|}{\nu}, \quad (2)$$

which is built from purely local quantities and tracks Re_θ remarkably closely in attached, two-dimensional boundary layers; the peak of Re_Ω remains within a few percent of $2.2 Re_\theta$ across the entire unstable Blasius range. This local proxy is the basis of the production term:

$$T(\tilde{\nu}, \nabla u) = f_\lambda(Re_{\text{esc}}, Re_\Omega) \Omega \tilde{\nu}, \quad f_\lambda = \frac{0.007 \max(0, Re_\Omega - 400)}{400 + 0.035 (Re_\Omega - 400)}. \quad (3)$$

The constant 0.007 is fixed analytically by requiring (3) to reproduce the canonical e^N envelope rate when integrated through the wall-normal direction of a Blasius profile (Fig. 1, right panel).

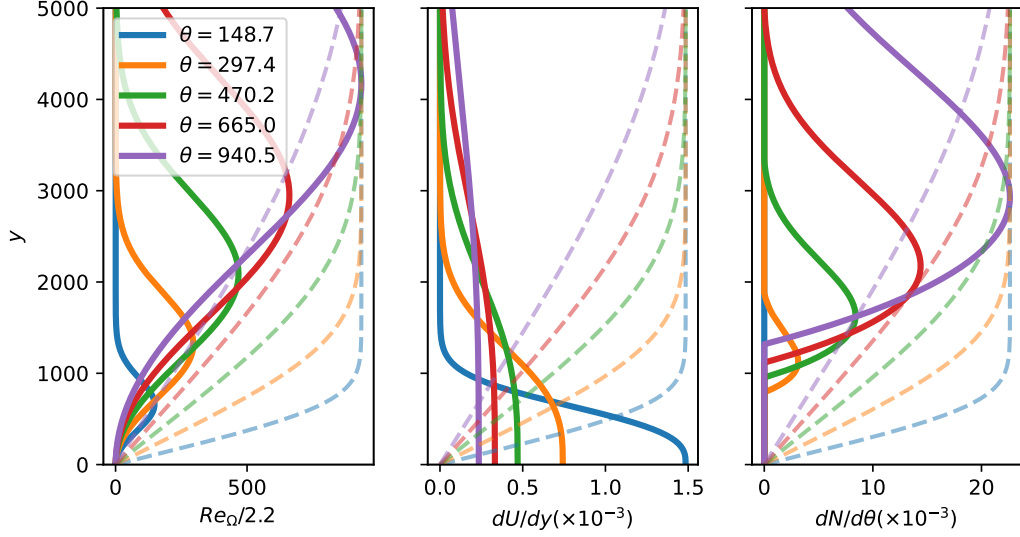


Figure 1: Calibration of Re_Ω -based growth in a Blasius layer. *Left*: streamwise profiles of u (solid) and Re_Ω (dashed); the peak of Re_Ω tracks $2.2 Re_\theta$. *Middle*: mean shear $\partial u/\partial y$ entering (3). *Right*: effective dN/dRe_θ from the local production integrated wall-normally; the model recovers the textbook e^N growth rate of 0.007 once $Re_\Omega^{\max} \gtrsim 1200$.

2.3 An integral pressure-gradient sensor: Re_{esc}

Real boundary layers are not Blasius. The e^N envelope rate dN/dRe_θ varies by an order of magnitude with shape factor: favorable gradients suppress amplification, adverse gradients accelerate it, and approaching separation triggers the Kelvin–Helmholtz mechanism. Existing local closures (Langtry–Menter, Coder–Maughmer) sense this through $\lambda_\theta \propto \partial v/\partial y$, which is brittle because v is small and its derivative is sensitive to mass-imbalance noise. We instead introduce

$$Re_{\text{esc}} \equiv \frac{d D d}{\nu D t} \approx \frac{y v}{\nu}, \quad (4)$$

the *Escape Reynolds number*: the wall-normal integral of the same quantity $\partial v/\partial y$ that traditional sensors differentiate. Dimensional analysis on Falkner–Skan similarity solutions confirms that Re_{esc} is independent of Re_x and depends primarily on the Hartree parameter β — it is small in stagnation flow, ~ 2 in Blasius flow, and grows past 4 as separation approaches. Its integral character makes it numerically smooth in unstructured RANS fields where the direct derivative is noisy.

We then close (3) into a 2D map $f_\lambda(Re_{\text{esc}}, Re_\Omega)$ with three asymptotic branches: (i) the viscous TS envelope ($1.4 < Re_{\text{esc}} < 3$), targeting $\lambda \approx 0.0025$; (ii) the inviscid shear limit ($Re_{\text{esc}} > 20$ or $Re_\Omega \gg 10^4$), targeting Michalke’s hyperbolic-tangent maximum $\lambda_{\text{max}} \approx 0.19$; and (iii) a linear blend in between. A simple gate $Re_\Omega > 100$ keeps the production term dormant in irrotational regions of the freestream.

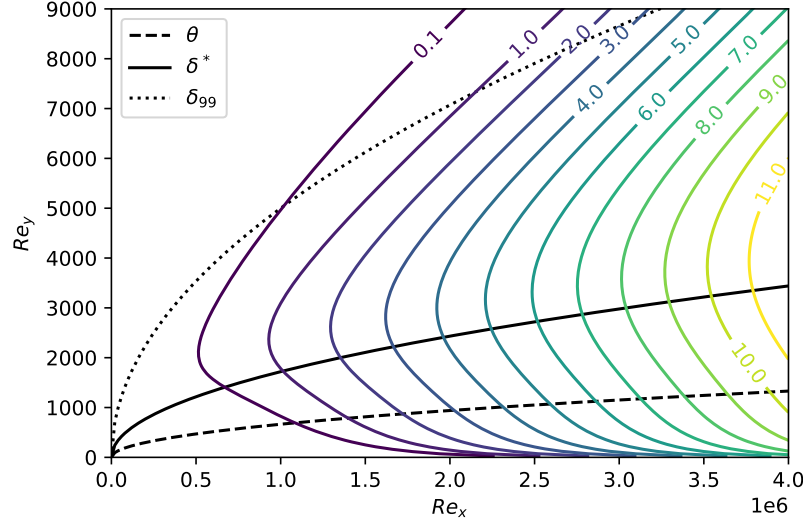


Figure 2: Solution of the full transport equation for $\hat{v}$ in a Blasius layer with uniform inlet $\hat{v}(0, y) = 1$. Contours of $N = \ln \hat{v}$ (0.1–11) remain confined to the TS band marked by θ , δ^* , and δ_{99} , demonstrating that local production — not diffusion — drives the streamwise march of transition.

3 Preliminary Results

3.1 Calibration on a Blasius boundary layer

Figure 1 shows that the calibration of (3) reproduces the classical envelope rate exactly. A finite-volume integration of the full transport equation, using $\hat{v}(x=0, y) \equiv 1$ as the inlet, then yields the N -factor field of Fig. 2: contours of $N = \ln \hat{v}$ remain confined to the TS-active band between θ and δ_{99} and grow many e -folds before cross-stream diffusion can homogenize the envelope, in quantitative agreement with e^N predictions. This confirms the design principle that the local production term, not the diffusion, sets the transition location.

3.2 Flat-plate parameter sweeps

A coupled boundary-layer/AFT-SA solver in the companion code base (`src/solvers/boundary_layer_solvers.py`) was used to sweep the inlet disturbance amplitude (a stand-in for freestream turbulence intensity) and verify the transition-onset trend. Figure 3 shows the resulting $C_f(Re_x)$ and $H(Re_x)$ envelopes; the model produces the qualitatively correct Mack- N_{crit} shift of transition location with disturbance amplitude.

3.3 2D RANS implementation

The closure is implemented in a JAX-accelerated 2D structured-grid RANS solver based on the Artificial Compressibility Method, JST convective fluxes, and a 5-stage Runge-Kutta time integrator. The solver shares its discrete operators between the SA branch and the AFT branch; switching between the two costs only an additional algebraic evaluation of f_λ inside the existing `compute_sa_source_jax` kernel. The end-to-end NACA 0012 case at $Re = 1 \times 10^6$, $\alpha = 0^\circ$ has been brought to converged solution and is being used for the validation campaign described below.

4 Expected Final Results

The final SciTech paper will deliver three categories of validation:

1. **Canonical flat plate.** Schubauer–Klebanoff and ERCOFTAC-T3 test cases at multiple turbulence intensities and pressure gradients, comparing transition location and skin-friction recovery against experiments and the SA-BCM and Langtry–Menter γ - Re_θ models.
2. **2D airfoils.** NACA 0012 and NLF 0416 over an angle-of-attack sweep at $Re = 10^6$ – 9×10^6 , comparing C_p , C_f , transition location, and integrated C_l , C_d against XFOIL (the reference e^N -based prediction at low freestream disturbance) and against correlation-based RANS transition models.
3. **Robustness in off-design conditions.** Cases with mild trailing-edge separation and laminar separation bubbles, where correlation-based models are known to be brittle; the goal is to demonstrate that the Re_{esc} -based switch into the inviscid (KH) branch prevents unphysical persistence of laminar separation, without requiring case-by-case retuning.

5 Significance for Aerospace CFD

If successful, the AFT-SA closure will collapse what is currently a two- or three-equation transition modeling problem onto a single transport equation that is structurally identical to SA. For aerospace solvers that already linearize SA implicitly (e.g. in JFNK or Newton–Krylov frameworks), adoption requires only an algebraic modification of the source term — no new conservation variables, no new boundary conditions, no new Jacobian blocks. This has the potential to make modern transition prediction accessible to industrial design workflows that currently default to fully turbulent SA out of implementation cost, while preserving the robustness that has made SA the workhorse closure of aerospace CFD for three decades.

Acknowledgments

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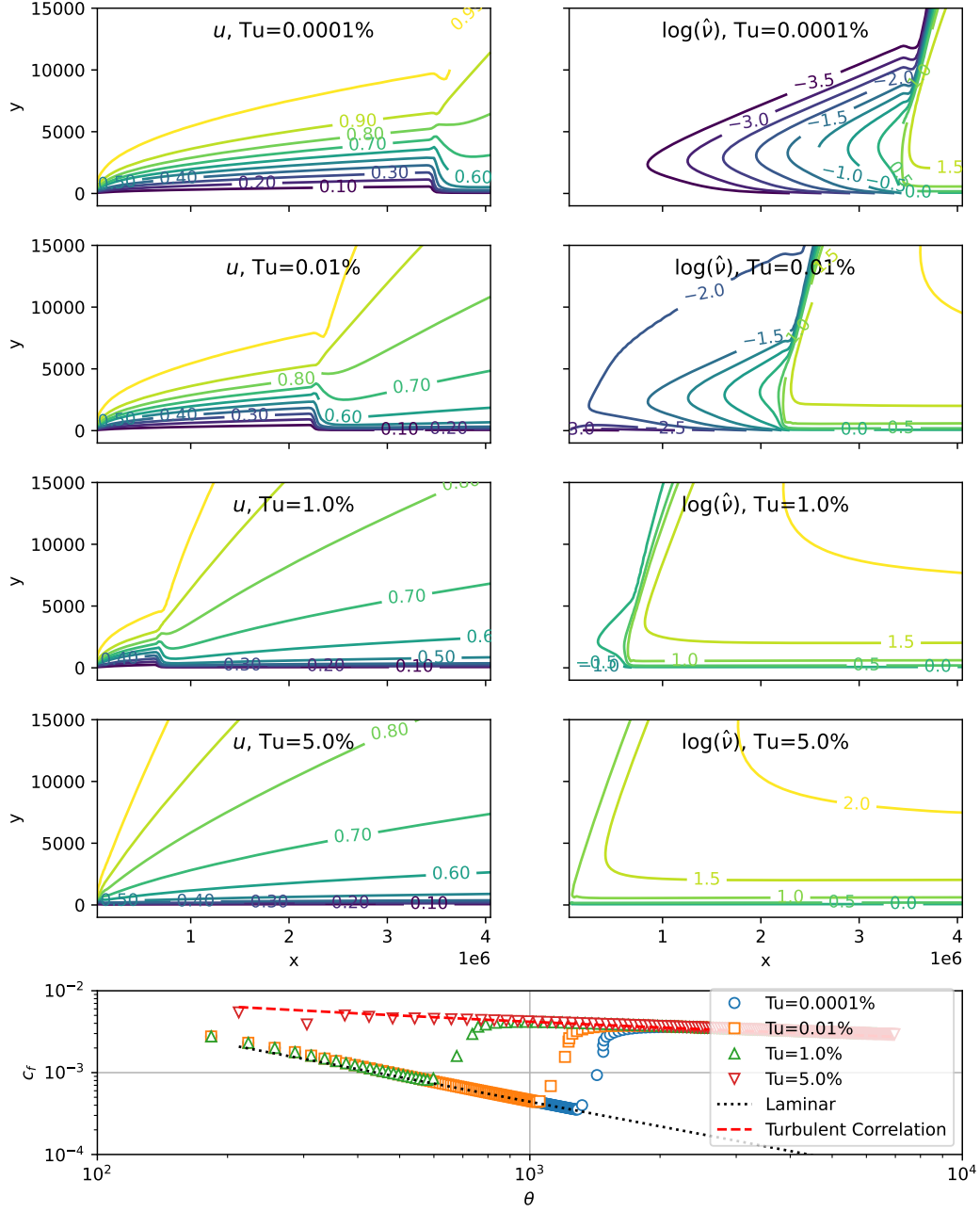


Figure 3: Flat-plate transition predictions across a range of inlet amplitudes (analogous to freestream turbulence levels). Skin friction (top) and shape factor (bottom) trace the canonical laminar-to-turbulent transition path; transition Re_x shifts upstream with disturbance amplitude in agreement with the Mack- N_{crit} trend.